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STL file format

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AI Overview

An STL (Stereolithography) file is a standard 3D file format that represents the surface geometry of a three-dimensional object as an unstructured mesh of connected triangles. Wikipedia +1

Key Characteristics

- **Surface Tessellation:** The entire outer shape of the 3D model is broken down (tessellated) into small triangular facets. www.fabbers.com +1
- **No Color or Texture:** STL files store only shape geometry and do not support color, materials, textures, or extra metadata like author and copyright information. Adobe +1
- **No Scale or Units:** The format does not record measurement units (such as millimeters or inches), which can occasionally cause models to open at the wrong size in slicer software. YouTube · 3D Printing P… +2
- **Two File Variants:**
  - **Binary STL:** Compact, fast to load, and widely used for actual 3D printing workflows.
  - **ASCII STL:** Human-readable plain text, larger in size, and mainly used for debugging or testing CAD interfaces. All3DP +2

What is it Used For?

- **3D Printing & Slicing:** Slicer software reads STL files to slice the 3D mesh into horizontal layers and convert them into G-code instructions for a printer. YouTube · Austin's Lab +2
- **Rapid Prototyping & CAM:** It serves as a universal bridge format between computer-aided design (CAD) software and manufacturing machines. Wikipedia +1

If you'd like, let me know:

- What **software or CAD program** you are using to export the file.
- What kind of **3D printer or project** you are preparing for.

I can share tips on adjusting mesh resolution or choosing the right export settings.

Ask anything

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STL (file format)

STL is a file format native to the stereolithography CAD software created by 3D Systems. Chuck Hull, the inventor of stereolithography and 3D Systems' ... Read more

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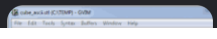
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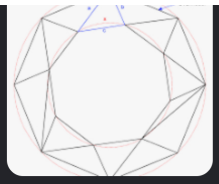
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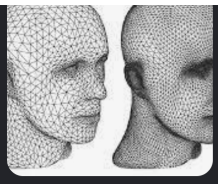
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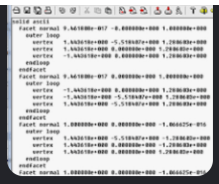





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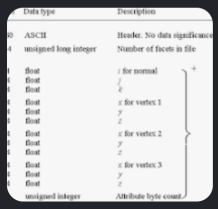
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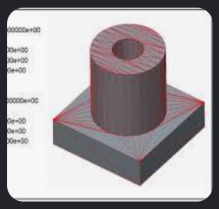
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STL File Format

The STL file format is **commonly used for 3D printing and rapid prototyping**, storing geometric surface data as a collection of triangular facets.

File Format Docs
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STL File

What is **STL** file? STL, abbreviation for stereolithrography, is an interchangeable file format that represents 3-dimensional surface geometry. Read more

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